

STAT 408 Lab 4: Putting it all together

Do a short start-to-finish analysis of a data set, either your own data or the data provided in lab (`fish_encounters.csv` or `relig_income.csv`). Then make a Quarto document in which you show your code for the following steps, using good style and code comments. **When you submit on Gradescope, select which pages correspond to each task.**

Read in the data.

- For this part, show the code you used to read in the data, but don't show output or show your dataset.
- In the text of the document, summarize whatever you had to do in order to read in the data properly, and why. Did you need to skip lines, rename columns, etc? Or was the data file already tidy enough that you were able to read it in without any arguments other than the file path?
- There should also be no package loading messages anywhere in the document you turn in.

Use some of the functions you know of to explore the data: how many rows and columns it has, what variables are in the dataset, whether they are quantitative or qualitative, continuous, categorical, etc. Do the variables have unexpected or missing values? What else might you want to know about your data before you start to work with it?

- Don't include the code for this part in your Quarto document.
- Summarize your observations about the data in the text of the document.

Compute some summary statistics of one or more columns of the data.

- Include the code in your report, as well as a summary of what you computed, why, and what you found.
- If the output of your summary statistics code is less than a page in the rendered pdf, show the full output in your report. Otherwise, show just the first few lines of the output with `head`.

Decide on a plot you think makes sense to make with this data, and make that plot. You may need to manipulate the data first (pivoting, making new columns, etc.) in order to make the plot of your choice.

- Show your code in the report.
- Summarize what you plotted and why, and what the plot tells you.

Finally, describe what else would be interesting to do or analyze with this data. Include at least one task that you tried to do with the data (or want to do with the data) that you don't yet have the skills to do.